

Set the Clock: Temporal Alignment of Pretrained Language Models

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Abstract

Language models (LMs) are trained on web text originating from many points in time and, in general, without any explicit temporal grounding. This work investigates the *temporal chaos* of pretrained LMs and explores various methods to align their internal knowledge to a target time, which we call “temporal alignment.” To do this, we first automatically construct a dataset containing 20K time-sensitive questions and their answers for each year from 2000 to 2023. Based on this dataset, we empirically show that pretrained LMs (e.g., LLaMa2), despite having a recent pretraining cutoff (e.g., 2022), mostly answer questions using earlier knowledge (e.g., in 2019). We then develop several methods, from prompting to finetuning, to align LMs to use their most recent knowledge when answering questions, and investigate various factors in this alignment. Our experiments demonstrate that aligning LLaMa2 to the year 2022 can enhance its performance by up to 62% according to that year’s answers. This improvement occurs even without explicitly mentioning time information, indicating the possibility of aligning models’ internal sense of time after pretraining. Finally, we find that alignment to a historical time is also possible, with up to $2.8\times$ the performance of the unaligned LM in 2010 if finetuning models to that year. These findings hint at the sophistication of LMs’ internal knowledge organization and the necessity of tuning them properly.¹

1 Introduction

Large-scale pretraining (Devlin et al., 2019; Raffel et al., 2020; Brown et al., 2020, i.a.) has enabled language models (LMs) to learn extensive knowledge from unlabeled text (Petroni et al., 2019). However, since pretraining corpora are constructed over a wide time period, they inevitably contain

outdated and contradictory information (Longpre et al., 2023). Previous studies have found that the temporal misalignment between LM pretraining and deployment has a significant impact on models’ performance (Lazaridou et al., 2021; Agarwal and Nenkova, 2022; Luu et al., 2022), which motivates many studies on making model’s knowledge up-to-date, by either continual learning (Jin et al., 2022; Ke et al., 2023), knowledge editing (Mitchell et al., 2022; Meng et al., 2023), or retrieval augmentation (Zhang and Choi, 2023; Vu et al., 2023). These approaches mainly focus on updating models with new knowledge and do not evaluate LM’s internal temporal knowledge across time.

In this work, we hypothesize that LMs, after pretraining, encode a chaotic sense of time, which means that models do not know which time-sensitive knowledge to use even if they have seen it during pretraining. We empirically investigate this temporal chaos and explore methods to do *temporal alignment*, which aims to align models’ internal knowledge to a target time.

We first introduce our TemporalAlignmentQA dataset (TAQA, §2) consisting of 20,148 questions, each with at least five different answers between 2000 and 2023. This dataset is automatically constructed based on Wikipedia tables that contain temporal information, making it easily scalable for future study as the world keeps changing. Complementary to previous temporal QA datasets (Chen et al., 2021; Dhingra et al., 2022a; Kasai et al., 2022; Liska et al., 2022; Tan et al., 2023; Wei et al., 2023), TAQA focuses on facts that changed multiple times during a recent period when most pretraining data is collected, so that it can be used to probe the temporal knowledge distribution of pretrained LMs. Figure 1 plots the F1 score of several representative LMs relative to ground-truth answers from each year, demonstrating that they tend to use earlier knowledge to answer questions, even when they have a very recent pretraining cutoff date (e.g.,

*Equal contribution.

¹Our dataset and code is available at <https://github.com/yizhongw/llm-temporal-alignment>.

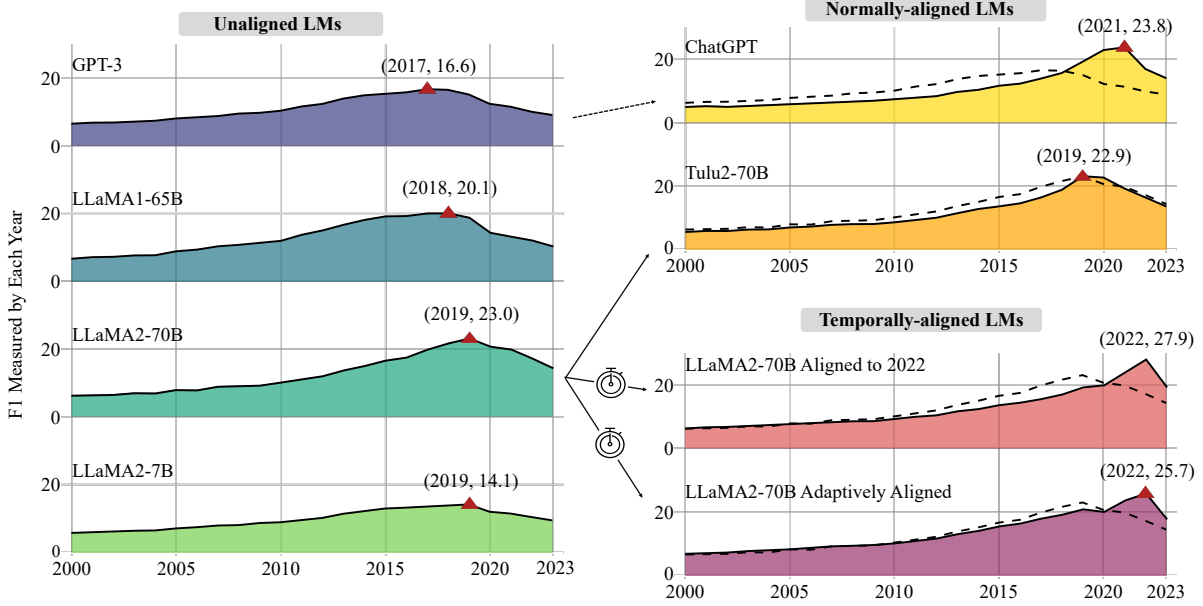


Figure 1: Performance (F1 score) of various LMs on our TAQA dataset, by year. Unaligned LMs (left) and conventionally aligned models (upper right) show relatively stronger performance when measured by the answers in earlier years, with their predictions more scattered across time. Our temporal alignment methods (lower right) lead to improved performance closer to a recent time (here, 2022) with a higher peak. The dotted line between GPT-3 and ChatGPT implies an uncertain relation (the latter is not necessarily derived from the former).

performance peaks in 2019 for LLaMa2, which has a cutoff of September 2022; [Touvron et al., 2023b](#)).

Next, we explore how to align LMs to answer questions based on a target time. Specifically, we first try aligning models to the most recent possible time, as given by their knowledge cutoff date (§3). This is often desired when LMs are deployed for general user populations who seek current answers to their questions. We propose three methods, including 1) *time-aware prompting*, where we prompt LMs using time information and time-sensitive examples; 2) *target-year finetuning*, where we finetune LMs with ground truth answers in a target year so that they can generalize and answer new questions based on that year; 3) *adaptive finetuning*, where we elicit the most recent knowledge each LM knows about a question (which can be earlier if the model does not know the updated version), and teach the model to answer adaptively based on the year proper for its own.

Experimental results (§4) show that finetuning LLaMa2 on TAQA, even without explicit temporal information, can relatively improve the answer performance in 2022 by up to 62.2%. We additionally explore the possibility of aligning LMs to a historical time (§5) and find our finetuning alignment strategy can boost the QA performance by 2.8× when aligning to 2010. We also find that 2019 is the most readily alignable year for LLaMa2, while

aligning it to 2015 causes the fewest errors measured against all valid answers from 2000 to 2023.

2 TAQA: A QA Dataset for Studying LM Temporal Alignment

To study the temporal chaos of pretrained LMs and how to align them temporally, we curate the TAQA dataset, featuring questions with frequently changing answers in a recent time period (2000–2023). We introduce the formulation of TAQA in §2.1, detail the data construction process in §2.2, and further describe the evaluation metrics in §2.3.

2.1 Problem Formulation

We formally define the task of TAQA as follows: given a question q , which has a set of answers $A = \{(a, t_s, t_e)\}$, where each answer a is only correct when the question is being asked during times $t \in [t^s, t^e]$. Such a dataset can be represented as $\mathcal{D} = \{(q_i, A_i)\}_{i=1}^N$, where at a specific timepoint t , the valid answers of question q shall be $\mathcal{A}(q, t) =$

$$\{a \mid \exists (a, t^s, t^e) \in A \wedge (q, A) \in \mathcal{D}, \text{ s.t. } t \in [t^s, t^e]\} \quad (1)$$

We consider a response from a QA system $\hat{a}$ to question q at time t as correct as long as $\hat{a} \in \mathcal{A}(q, t)$. Since only questions with varied answers through history are useful to us, we quantify a question’s time-sensitivity as the number of unique an-

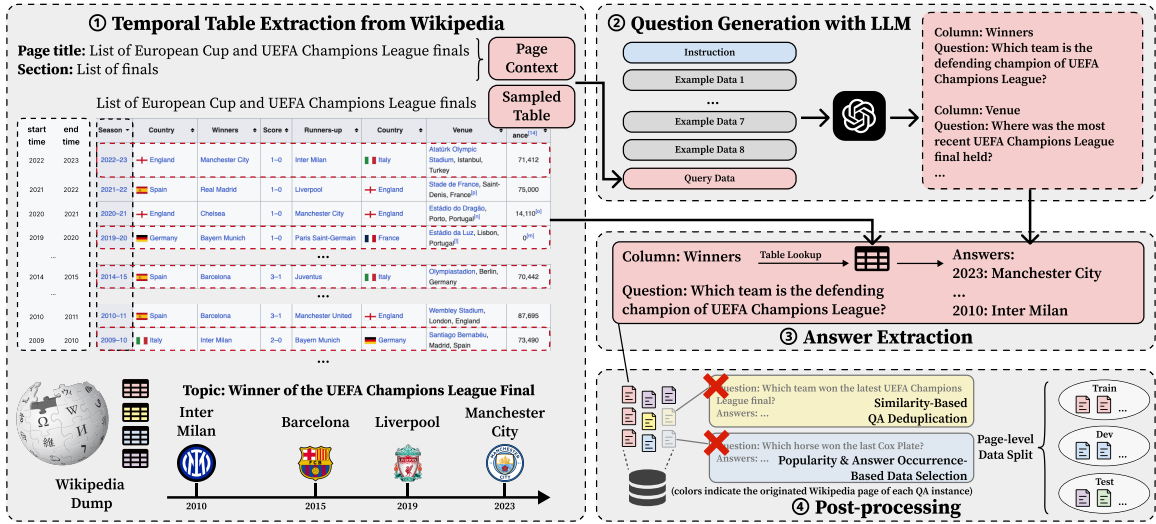


Figure 2: The data construction process of our TAQA dataset.

swers during a time, i.e.,

$$S(q, t_s, t_e) = |\{A(q, t) \mid t \in [t_s, t_e]\}| \quad (2)$$

In practice, we only keep questions such that $S(q, 2000, 2023) \geq 5$ in our TAQA dataset.

2.2 Data Construction

To get question q with its multiple temporal answers (a, t_s, t_e) , we use tables with temporal columns from the Wikipedia dump to build our TAQA dataset, as the temporal columns can be seen as qualifiers t_s, t_e and other columns can be leveraged to generate QA pairs. The advantage of Wikipedia is that its topics are popular and well-structured. Also, the topic distribution of Wikipedia is less biased than open knowledge graphs, such as Wikidata (Piscopo et al., 2017). Furthermore, Wikipedia is used in the pretraining corpus for most modern LMs, ensuring that LMs have learned such knowledge from pretraining. We demonstrate our steps for constructing the TAQA dataset as shown in Figure 2 as follows.

Temporal table extraction from Wikipedia.

Given the Wikipedia dump,² we use the WikiExtractor (Attardi, 2015) to extract all tables from the dump into the CSV format. Then, we use heuristic-based methods to identify the columns denoting temporal information (i.e., (t_s, t_e)) for answers. As we want to gather questions for contemporary knowledge only, we only keep tables with information for every year from 2010 to 2023, resulting in 17,932 tables.

Question generation with LMs. To generate natural questions with varied styles, we prompt GPT-4 with our manually curated few-shot examples with the tables gathered. We also add each table’s Wikipedia page information (title and section names) to the prompt to provide sufficient context. To reduce the query cost, we sample the table and use only the rows corresponding to 2010, 2020, and 2023. To extract answers from the table later, we instruct GPT-4 to generate the column name before each question. After this process, 96,309 question-column pairs are generated in total. Appendix §C presents the detailed prompt.

Answer extraction. With the question-column pairs generated, we use the column names to extract the answers to curate QA pairs from the table directly. When multiple rows have a shared period $[t_s, t_e]$, we treat all answers in those rows as valid. To ensure the question q aligns with our sensitivity requirement $S(q, 2000, 2023) \geq 5$, we only keep the QA pairs with at least five unique answers.

QA postprocessing. To avoid data leakage in our TAQA dataset, we discard QA instances based on their question and answer similarities, measured using BM 25 scores calculated with Lin et al. (2021). When duplicates are found, we keep the instance with fewer words in the question as they are often more natural and popular among real people. Next, we attempt to reduce the dataset’s answer bias as we do not want LMs to improve their QA performance by memorizing frequently occurring answers. We begin by identifying questions with answers that are numerical or occur more than 300 times and randomly sample only 10% of them to

²We use the English Wikipedia dump of January 1, 2024.